

Simple symmetric Venn diagrams with 17 curves

Chris Dzoba

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Abstract

We exhibit simple, rotationally symmetric Venn diagrams with 17 curves, the smallest open case since the 11- and 13-curve diagrams of Mamakani and Ruskey (2014). Four such diagrams were found by a Metropolis walk on rotation-invariant quadrangulations of the sphere in which regions may temporarily be duplicated. Each diagram is given by a machine-checkable certificate. We also exhibit the first non-monotone simple symmetric Venn diagrams with 11 and 13 curves.

1 Statement

An n -Venn diagram is a family of n Jordan curves in the plane such that for every subset S of the curves the set of points inside exactly the curves of S is nonempty and connected. It is *simple* if no point lies on three curves, and *symmetric* if a rotation by $2\pi/n$ maps the family to itself. Symmetric Venn diagrams exist for every prime n (Griggs, Killian, Savage 2004), but simple ones were known only for $n \leq 13$.

Theorem 1. *Simple symmetric Venn diagrams with 17 curves exist.*

2 Encoding and verification

Let D be a simple symmetric n -Venn diagram, drawn on the sphere with the unbounded region as a face. Its dual map M has one vertex per region, labelled by the region's membership pattern in $\{0,1\}^n$; one edge per arc, joining labels that differ in exactly the bit of the arc's curve; and one face per crossing, a 4-cycle with labels $u, u \oplus e_a, u \oplus e_a \oplus e_b, u \oplus e_b$ (a *cube square*). Conversely:

Proposition 1. *A map on the sphere whose vertices are labelled by distinct elements of $\{0,1\}^n$, all 2^n occurring, whose faces are cube squares, whose edges of each coordinate i form a single cycle through the faces containing them, and whose vertex rotations are single cycles, is the dual of a simple n -Venn diagram; if the label rotation $x \mapsto (x_1, \dots, x_{n-1}, x_0)$ is an orientation-preserving automorphism of the map, the diagram is symmetric.*

Proof sketch. The map is a cellular embedding on the sphere; its geometric dual is a 4-regular plane graph whose vertices are the faces (crossings). The edges of coordinate i , chained through opposite sides of each crossing, form one closed curve C_i ; regions of the arrangement are the vertices of M , and by the label condition each point of $\{0,1\}^n$ is realized by exactly one region, which is a disk. Rotation of the labels acts as a rotation of the sphere fixing the two vertices 0^n and 1^n ; taking 0^n as the outer face gives a plane diagram with n -fold rotational symmetry. $\square$

A certificate is a list of the $2^{17} - 2 = 131,070$ faces of M , each as four 17-bit labels. Verification (seconds) recomputes: all labels distinct and present (2^{17} vertices), every edge in exactly two faces (262,140 edges), Euler characteristic 2, every face transversal, single rotation cycle at every vertex, one cycle per curve with both sides connected, and invariance under label rotation. Two independently written checkers (a half-edge implementation in C++ and a face-set implementation in Python) accept all four certificates, and the first certificate (`c178d7bd`) has been formally verified in Lean 4 with Mathlib by Justin Grimes, working from the definitions alone and without access to our code: his theorem states that the 17 curves constructed from the certificate are Jordan curves in the Euclidean plane, carried to one another by a rigid rotation through $2\pi/17$, with every one of the 2^{17} regions path-connected, no point on three curves, and every crossing a transversal crossing of two curves. The finite parts of the proof use Lean’s compiled evaluation (`native_decide`); the development builds from scratch in about nine minutes. SHA-256 prefixes of the certificate files: `c178d7bd`, `87f981b8`, `eec8cd55`, `431857aa`.

3 Method

We search the space of rotation-invariant cube-square quadrangulations of the sphere in which several vertices may carry the same label. The energy of a state is $E = (\text{missing labels}) + \lambda (\text{duplicate regions})$; $E = 0$ with $\lambda > 0$ is exactly a certificate. Moves are the local Reidemeister-type rewirings of the dual (lens insertion and removal on a 3-cube, triangle flip, bigon insertion and removal), each applied simultaneously to all n rotation images; every move preserves the sphere, the cube-square faces and the one-cycle-per-curve property, so the walk never leaves the space of valid symmetric arrangements. Proposals are accepted by the Metropolis rule at temperature T , with half of the proposals targeted at a missing label or a duplicated region.

Allowing duplicates is what makes the search work: with duplicates forbidden, greedy growth of the same quadrangulations reaches 98.3% of the labels and jams, because the last lens insertions require two missing labels to coincide geometrically. With duplicates allowed, a missing label can be created next to a duplicate at zero cost and defects diffuse until they annihilate.

The walk starts from the Griggs–Killian–Savage diagram for $n = 17$ with every multiple crossing resolved into simple crossings, a state with all labels present and 69,394 duplicate regions. With λ ramped from 0.3 to 1 over three hours at $T = 0.25$, three of fifteen independent runs reached $E = 0$; a fourth diagram was obtained by a further 40-minute cycle from a state with 68 duplicates. Runs at $T = 0.20$ reached $E = 34$ but did not finish. The same method finds 11- and 13-curve diagrams from the trivial starting state in seconds and minutes; every diagram it produced at $n = 11, 13, 17$ is non-monotone (some k -region lacks a neighbouring $(k-1)$ - or $(k+1)$ -region), whereas all previously published simple symmetric Venn diagrams with $n \geq 11$ are monotone.

4 Data

The certificates (about 11 MB each), an independent verification script, the search code and the plotter are at <https://github.com/dzoba/venn17> and archived on Zenodo (DOI to be inserted).

Contributions and use of AI systems

This work was carried out by two AI systems working under the author’s direction over three days, and this note discloses their roles in full. Claude (Anthropic) formulated the dual-map encoding, designed and implemented the search (the relaxed walk, its moves, energy and annealing schedules), ran the searches and the verification, performed the analyses reported here and drafted this note. Codex (OpenAI) constructed the resolved Griggs–Killian–Savage starting states, wrote independent validators for the certificates, identified the need for the bigon moves, and built exact local solvers used to diagnose the earlier plateau. The author set the goals and strategy, decided what to pursue and what to abandon, supplied the computing resources, and is responsible for the content.

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